



Department of Creative Technologies
Web Engineering & AI Lab

Course: Web Engineering and AI

Lab Topic: CSS

Level: Undergraduate

Lab Duration: 3 Hours

Software Required: VS Code (or any code editor), Web Browser (Chrome/Edge/Firefox)

Prepared by: Muhammad Abdullah

Cluster Head: Dr. Adnan Aslam

1. Lab Objectives

By the end of this lab, students should be able to:

1. Understand the purpose of CSS in web development.
2. Apply CSS using different methods.
3. Understand CSS selectors and properties.
4. Style HTML elements using colors, fonts, borders, margins, and padding.
5. Understand the CSS Box Model.
6. Create basic layouts using CSS.

Lab Instructions

Students are required to submit the following for each lab task:

1. Individual code files containing the source code for each task.
2. One Word or PDF file containing the complete solution of the lab manual.

This file must include:

1. Cover Page
2. Tasks/Exercises
3. Source Code
4. Output Screenshots
5. Outcome
6. The student's Name and Registration ID clearly mentioned at the beginning of the document.
7. The file must follow the required naming format <Reg ID>_<student name>_<course name>_<Lab No>, (for example: 221XXX_Muhammad Abdullah_Web_Lab_01)
8. Please ensure that all tasks are completed, properly labeled, and arranged according to the given lab manual and template. Plagiarized Content will get **Absolute 0**

Part A — CSS

1. Introduction to CSS

CSS stands for **Cascading Style Sheets**.

HTML defines the **structure** of a webpage, while CSS defines its **appearance and presentation**.

```
<h1>Welcome to Web Development</h1>
<p>This is my first webpage.</p>
```

Without CSS, the browser displays this content using its default styling. CSS lets us change text color, background color, font size and family, borders, width/height, spacing, alignment, layout, buttons, navigation bars, cards, and responsive behavior.

Simple Example

```
<h1 style="color: blue;">Hello Students</h1>
```

Here `color` is a CSS **property** and `blue` is its **value**.

General syntax:

```
selector {
  property: value;
}
```

```
h1 {  
  color: blue;  
  font-size: 30px;  
}
```

2. Ways to Apply CSS

2.1 Inline CSS

Written directly inside an HTML element's `style` attribute.

```
<h1 style="color: green; font-size: 35px;">Welcome</h1>
```

- **Advantage:** quick to apply to a single element.
- **Disadvantage:** hard to maintain across many elements.

2.2 Internal CSS

Written inside a `<style>` element in the `<head>`.

```
<!DOCTYPE html>  
<html>  
  <head>  
    <style>  
      h1 { color: blue; }  
      p { font-size: 18px; }  
    </style>  
  </head>  
  <body>  
    <h1>My Website</h1>  
    <p>This is my paragraph.</p>  
  </body>  
</html>
```

2.3 External CSS

Written in a separate `.css` file and linked to the page. Preferred for larger websites since one file can style multiple pages.

`index.html`

```
<head>
<link rel="stylesheet" href="style.css">
</head>
```

`style.css`

```
h1 {
  color: blue;
  font-size: 35px;
}
```

3. CSS Selectors

A selector tells CSS **which** HTML element(s) to style.

Element Selector — targets all elements of a type:

```
p { color: blue; }
```

ID Selector — targets one specific element (`#`):

```
<p id="message">Hello Students</p>

#message { color: red; }
```

Class Selector — reusable across multiple elements (`.`):

```
<p class="highlight">Paragraph 1</p>
<p class="highlight">Paragraph 2</p>

.highlight { color: green; }
```

Group Selector — style several elements together:

```
h1, h2, p { font-family: Arial; }
```

4. CSS Colors

```
p { color: red; } /* named color */
p { color: #ff0000; } /* hexadecimal */
p { color: rgb(255, 0, 0); } /* rgb */
body { background-color: lightgray; }
```

5. CSS Text and Fonts

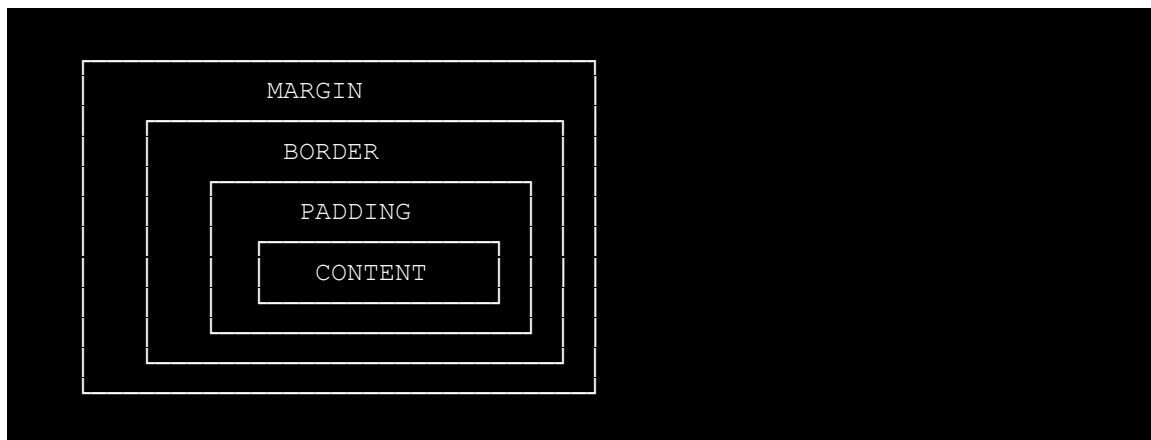
```
p {
  color: black;
  font-size: 18px;
  font-family: Arial;
  font-weight: bold;
  text-align: center;
}
```

Property	Purpose
color	Text color

Property	Purpose
font-size	Text size
font-family	Font type
font-weight	Boldness
text-align	Text alignment
text-decoration	Underline, etc.
line-height	Space between lines

6. CSS Box Model

Every HTML element is treated as a box made of four layers, from the inside out:



Content → Padding → Border → Margin

```
div {  
  width: 300px;  
  padding: 20px; /* space inside, between content and border */  
  border: 2px solid black;  
  margin: 30px; /* space outside the element */  
}
```

7. CSS Width and Height

```
.box {  
  width: 300px;  
  height: 200px;  
}  
.box {  
  width: 50%;    /* percentage of parent */  
}
```

8. Styling Buttons

```
<button>Click Me</button>
```

```
button {  
  background-color: blue;  
  color: white;  
  padding: 10px 20px;  
  border: none;  
  border-radius: 5px;  
}  
button:hover {  
  background-color: darkblue;  
}
```

9. InLab Task

1. **Create** an HTML page with a heading and a paragraph. Style them using **internal CSS** (choose your own colors, font, and alignment).
2. **Create** a `.css` file and link it externally to style a `div` with a background color, border, padding, and rounded corners.
3. **Create** a page with a heading, three paragraphs, an image, and a button. Use CSS to change colors and fonts, add borders and spacing, center the heading, style the button, and add a hover effect.